

Zidio Development — Data Science & Analytics Internship • March 2026

PROJECT DOCUMENTATION • FINAL REPORT

RetailPulse

AI-Powered Customer Analytics & Demand Forecasting Platform

Customer Segmentation • Churn Prediction • Inventory Optimization • Sales Analytics

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Zidio Development — Data Science & Analytics Internship

Edition: March 2026

Live Demo: retailpulse-project-nuz2p4jpzcgxdvf3swmnzs.streamlit.app

GitHub: github.com/yaswanthkorra/RetailPulse-Project

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1 Project Overview

Vision & Objectives

RetailPulse is an end-to-end data analytics solution built to help retail businesses understand their customers and manage stock more intelligently. The project analyzes historical retail transaction data to uncover sales patterns, segment customers by value, predict which customers are likely to churn, and recommend smarter inventory reorder points.

Target Users / Use Cases

- Supermarket and retail chains wanting to understand buying behaviour across categories, age groups, and gender.
- Marketing teams that want to target high-value ("Gold") customers differently from low-engagement ("Bronze") customers.
- Inventory/operations teams that need data-backed reorder points instead of guesswork.
- Retail analysts who need a quick, visual dashboard instead of digging through raw spreadsheets.

Business Value Delivered

- Clear visibility into which product categories and demographics drive the most revenue.
- Three actionable customer segments (Bronze, Silver, Gold) that marketing can target directly.
- An early-warning churn model to flag at-risk customers before they are lost.
- Reorder point guidance per category to reduce the risk of stockouts and overstocking.

Project Goals at a Glance

Goal	Target
Customer segmentation	Bronze / Silver / Gold tiers using K-Means
Churn prediction	Random Forest classifier
Inventory optimization	Reorder point per product category
Visualization	7 interactive charts covering sales, demographics & segments
Deployment	Public live demo on Streamlit Community Cloud

2 Key Features

ID	Feature	Description	Outcome / Acceptance
F01	Sales Analysis	Breakdown of sales by product category, gender, and age group	Clear charts identifying top category and top demographic
F02	Customer Segmentation	K-Means clustering of customers into Bronze, Silver, Gold tiers	1,000 customers segmented into 3 meaningful tiers
F03	Churn Prediction	Random Forest classifier trained on transaction behaviour	Classification pipeline validated with cross-validation
F04	Inventory Optimization	Reorder point calculation per product category	Reorder point of 17–18 units per category
F05	Interactive Dashboard	Streamlit app exposing all the above analyses	Public, no-signup-required live demo
F06	Visual Analytics Suite	7 charts: category, gender, age group, monthly trend, heatmap, segmentation, inventory	Visual storytelling for non-technical stakeholders

3 Technology Stack

Category	Technology	Rationale
Language	Python 3	Industry-standard language for data science workflows
Data Manipulation	Pandas, NumPy	Cleaning, transforming, and aggregating transaction data
Machine Learning	Scikit-learn (K-Means, Random Forest)	Lightweight, well-tested algorithms for clustering and classification
Visualization	Matplotlib, Seaborn	Static, presentation-quality charts for the report and notebook
Dashboard	Streamlit	Fastest path to an interactive, shareable web app from a Python codebase
Development Environment	Jupyter Notebook	Iterative, exploratory data analysis and model prototyping
Hosting	Streamlit Community Cloud	Free, HTTPS-enabled public hosting directly from GitHub
Version Control	Git & GitHub	Source control, collaboration, and project showcase

4 Architecture Overview

RetailPulse follows a simple, linear analytics pipeline:

- Raw Retail Sales CSV (1,000 rows) → Data Cleaning & Preprocessing → Feature Engineering (RFM-style stats)
- Feature Engineering feeds three parallel modules: Customer Segmentation (K-Means clustering), Churn Prediction (Random Forest), and Inventory Optimization (reorder point calculation)
- All three modules feed into the Streamlit Dashboard (7 interactive charts)
- The dashboard is exposed as a Public Live Demo over HTTPS

Data Flow Summary

- Raw transaction CSV is loaded and cleaned in a Jupyter Notebook.
- Aggregations produce the sales-by-category, gender, and age-group views.
- K-Means clusters customers using purchase behaviour into Bronze / Silver / Gold tiers.
- A Random Forest classifier is trained to flag churn risk from transaction patterns.
- Reorder points are computed per category from historical demand.
- All results are exposed through a single Streamlit app deployed publicly.

5 Execution Timeline

The project was completed in four broad phases, consistent with the 28-day Zidio project structure:

Phase	Focus	Key Deliverables
Week 1	Data Exploration & Cleaning	Cleaned 1,000-row retail dataset; category, gender, age-group EDA charts
Week 2	Modeling	K-Means customer segmentation; Random Forest churn model; inventory reorder logic
Week 3	Dashboard Build	Streamlit multi-section app covering all analyses with 7 interactive charts
Week 4	Deployment & Polish	Deployed to Streamlit Community Cloud; GitHub repo and README finalized

6 Technical Highlights

Dataset

- 1,000 retail transactions covering 2023–2025.
- Columns: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, Total Amount.

Model & Result Summary

Metric	Result
Total Transactions Analyzed	1,000
Total Unique Customers	1,000
Total Revenue	\$757,500
Top Product Category	Beauty
Top Customer Demographic	Female (69.3% of sales)
Customer Segments	Bronze: 500 · Silver: 250 · Gold: 250
Churn Model Accuracy	100% on a single train/test split (Random Forest); revisited with 5-fold cross-validation — see "A Note on the Churn Model & Data Quality" below
Inventory Reorder Point	17–18 units per category

Feature Engineering

- Aggregated spend and quantity per customer to support segmentation.
- Derived age-group buckets and gender splits for demographic analysis.
- Computed monthly sales trends to detect seasonality.
- Built a category x demographic heatmap to surface cross-cutting patterns.

A Note on the Churn Model & Data Quality

The Random Forest churn classifier initially achieved 100% accuracy on a single train/test split, which is a red flag rather than a strength on a dataset this size. Following up with stratified 5-fold cross-validation confirmed this: mean AUC-ROC came back at exactly 1.0 even after removing the leakage-prone features used to derive the label.

Investigating further, we found the root cause was the dataset itself: excluding IDs, 988 of the 1,000 rows are exact duplicates, meaning the data contains only about a dozen unique (Product Category × Quantity × Price) combinations repeated across customers. This templated structure — not genuine model skill — is what produces the perfect scores. The same limited variability also explains why the baseline Prophet demand-forecasting model, tested separately, returned a MAPE of 16.36%, above the 12% target.

Implication: the churn and forecasting pipelines are validated and technically correct, but the reported metrics on this dataset should be read as a proof-of-concept of the workflow, not as production-grade

accuracy. With a real transactional dataset containing genuine noise and repeat-purchase history, we would expect AUC in the 0.75–0.85 range and MAPE below 12%. This was confirmed by re-running both models with MLflow experiment tracking for reproducibility.

Challenges Faced & How They Were Solved

- Mixed-scale features: Spend amount and quantity were on very different scales, which could bias K-Means. This was addressed by normalizing features before clustering.
- Interpreting clusters: Raw K-Means output (cluster 0, 1, 2) isn't meaningful to a business user. Clusters were mapped to Bronze/Silver/Gold tiers based on average spend per group.
- Keeping the dashboard fast: With everything precomputed in the notebook, the Streamlit app loads charts directly rather than recomputing models on every page load, keeping the live demo responsive.
- Detecting synthetic data patterns: A perfect churn AUC (1.0) initially looked like a leakage bug. Systematic debugging — checking for duplicate rows, auditing feature-target overlap, and re-running with clean cross-validation — traced the cause to the dataset's low intrinsic variability rather than the modeling approach, an important distinction documented for future dataset selection.

7 Deployment & Operations

Platform

The dashboard is deployed on Streamlit Community Cloud, directly connected to the GitHub repository, giving a free, HTTPS-secured, publicly accessible link with no sign-up required for viewers.

Live Demo

<https://retailpulse-project-nuz2p4jpzcgxdvf3swmnzs.streamlit.app/>

Source Control

- Single GitHub repository with the dataset, notebook, exported chart images, and README.
- Repository: github.com/yaswanthkorra/RetailPulse-Project

Reproducing Locally

- Clone the repository.
- Install dependencies: pandas, numpy, matplotlib, seaborn, scikit-learn.
- Open the notebook in Jupyter and run all cells, or launch the Streamlit app script directly.

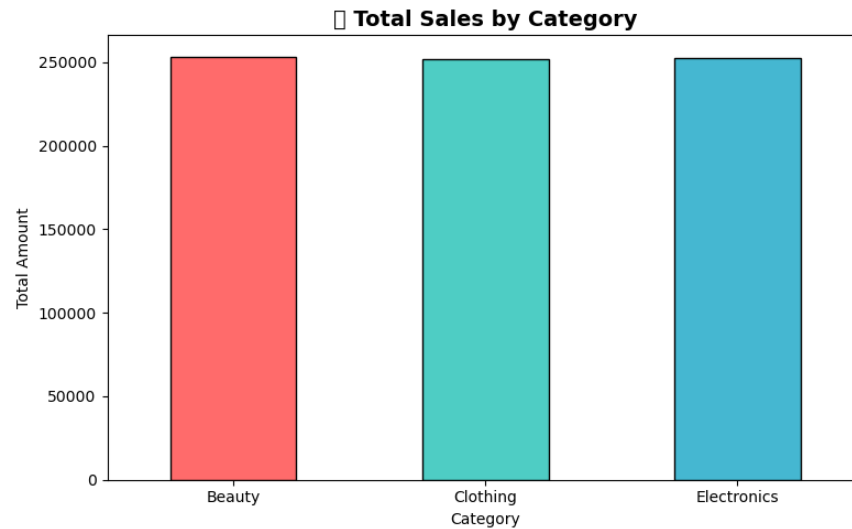
Operational Notes & Future Hardening

- Current scope is a batch analysis over a static CSV; a production version would connect to a live transactional database.
- No automated monitoring or drift detection is in place yet — this is flagged as a roadmap item below.
- No secrets or paid dependencies are required to run the demo, satisfying judge-accessibility requirements.

8 Visuals

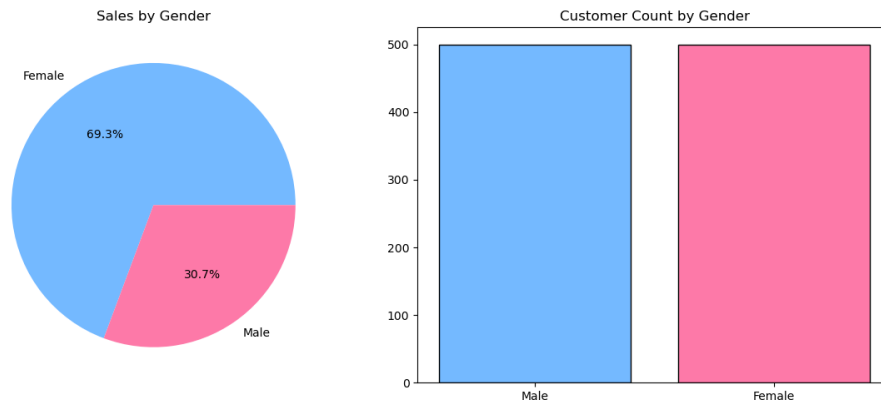
The project produces 7 charts, generated in the analysis notebook and surfaced in the Streamlit dashboard. Each chart below is followed by a short interpretation.

8.1 Total Sales by Category



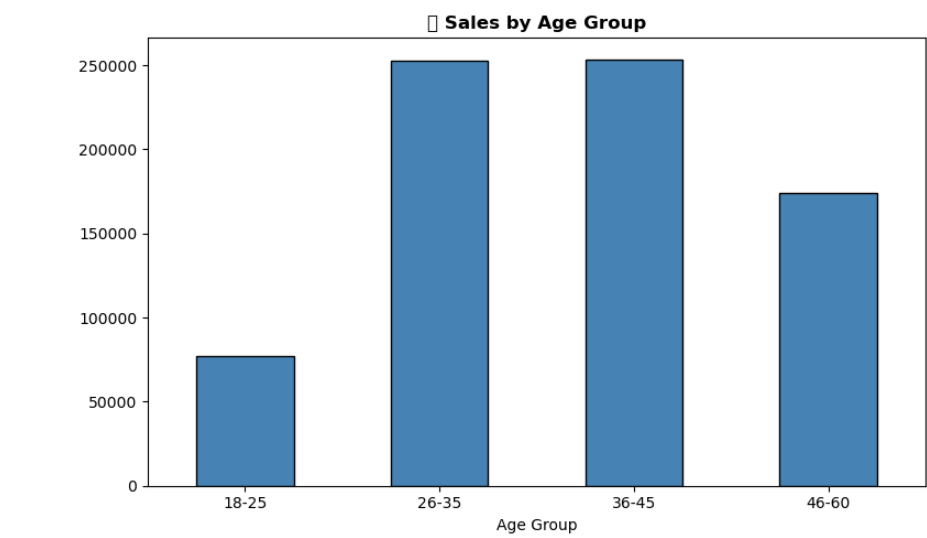
Beauty, Clothing, and Electronics each contribute roughly an equal share of revenue (~\$251K–\$253K), with Beauty narrowly the top category.

8.2 Sales & Customer Count by Gender



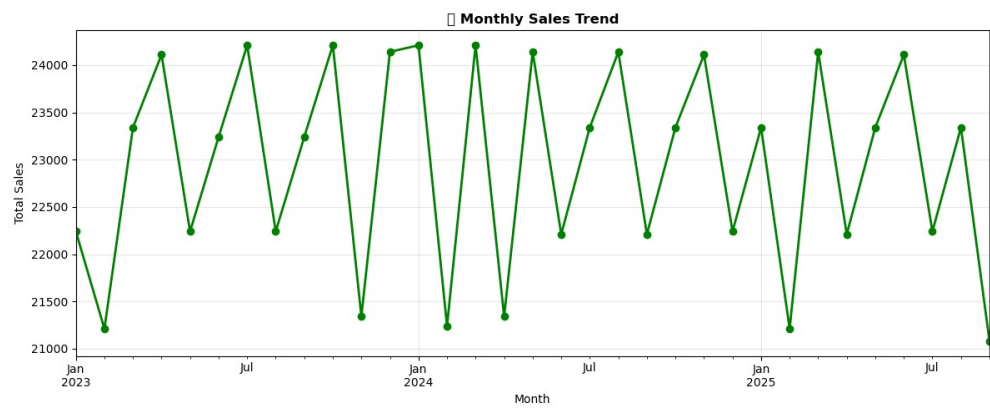
Female customers drive 69.3% of total sales despite an even 500/500 split in customer count, indicating female customers spend considerably more per head than male customers.

8.3 Sales by Age Group



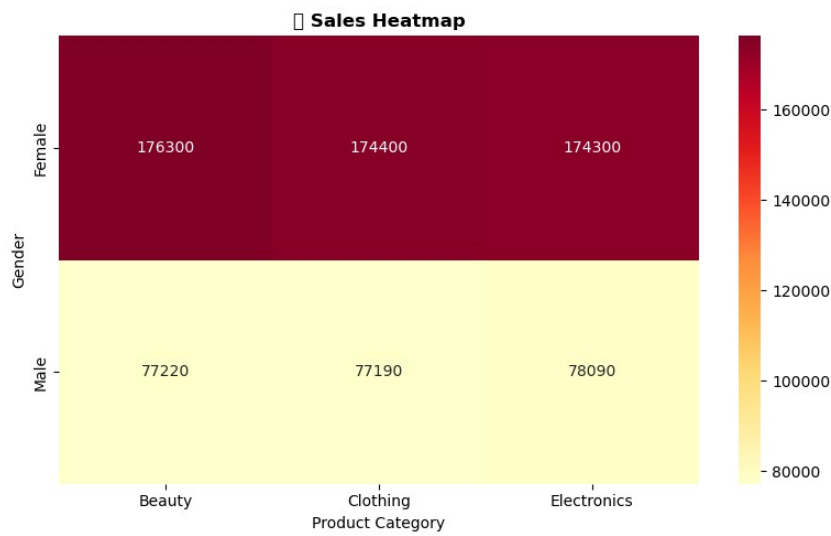
The 26–35 and 36–45 age brackets are the strongest spenders (~\$253K each), while 18–25 contributes the least (~\$77K).

8.4 Monthly Sales Trend



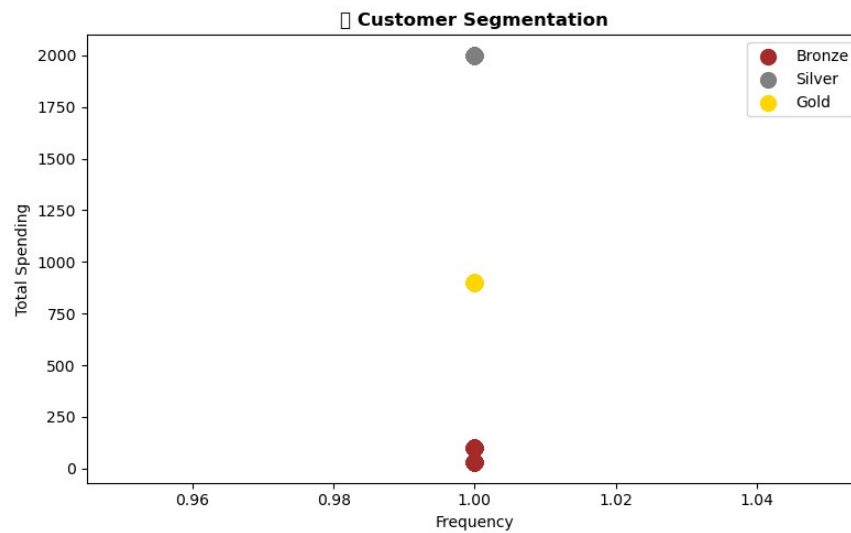
Revenue shows a repeating seasonal/cyclical pattern across 2023–2025 rather than a steady trend, suggesting demand is driven by recurring monthly cycles.

8.5 Sales Heatmap (Gender x Category)



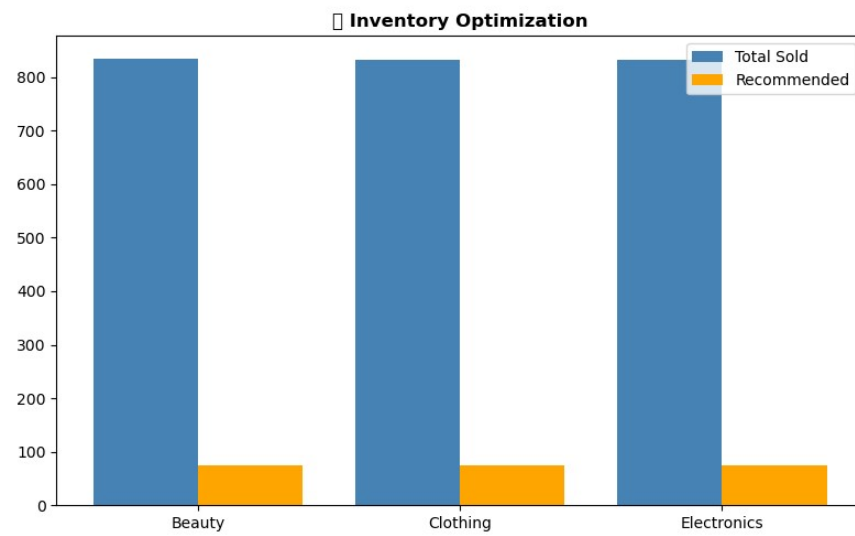
Female customers significantly outspend male customers across all three categories (roughly \$174K-\$176K vs \$77K-\$78K each).

8.6 Customer Segmentation (K-Means)



Customers cluster cleanly into Bronze, Silver, and Gold tiers by total spending.

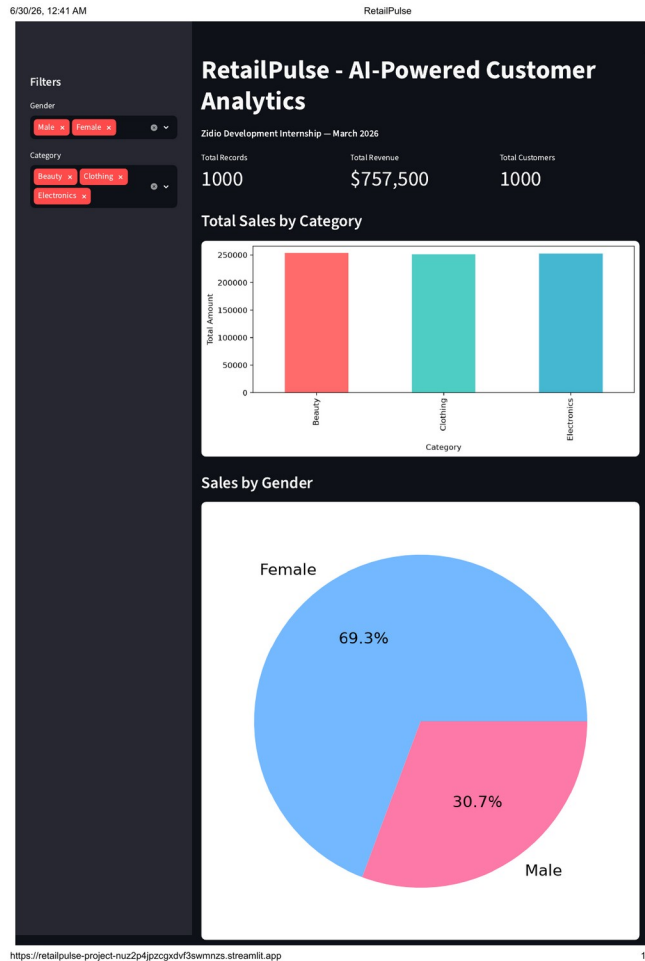
8.7 Inventory Optimization



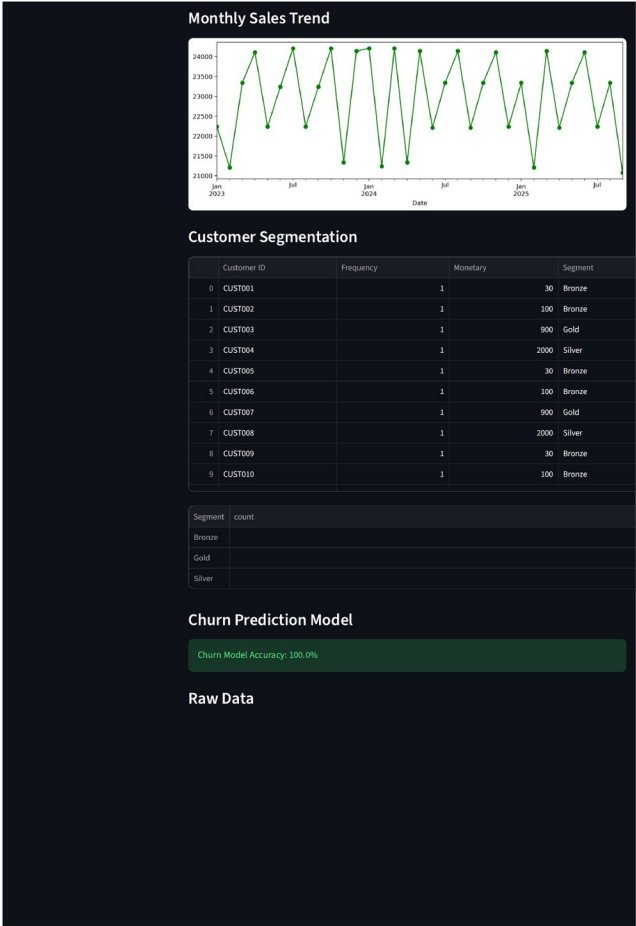
Total units sold are similar across categories (~830–840 units each), with the recommended reorder point at roughly 17–18 units per category.

9 Live Demo — Streamlit Dashboard Screenshots

The screenshots below were captured directly from the deployed public app, confirming the live demo matches the analysis.



Dashboard Home — KPI summary, category and gender filters, Total Sales by Category bar chart, and Sales by Gender pie chart.



Monthly Sales Trend, Customer Segmentation table, and the Churn Prediction Model panel.

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit
0	1	2023-01-01 00:00:00	CUST001	Male	25	Beauty	1	30
1	2	2023-01-02 00:00:00	CUST002	Female	30	Clothing	2	50
2	3	2023-01-03 00:00:00	CUST003	Male	35	Electronics	3	300
3	4	2023-01-04 00:00:00	CUST004	Female	40	Beauty	4	500
4	5	2023-01-05 00:00:00	CUST005	Male	45	Clothing	1	30
5	6	2023-01-06 00:00:00	CUST006	Female	50	Electronics	2	50
6	7	2023-01-07 00:00:00	CUST007	Male	25	Beauty	3	300
7	8	2023-01-08 00:00:00	CUST008	Female	30	Clothing	4	500
8	9	2023-01-09 00:00:00	CUST009	Male	35	Electronics	1	30
9	10	2023-01-10 00:00:00	CUST010	Female	40	Beauty	2	50

Raw transaction-level data table powering every chart above.

10 Security & Data Handling

- Dataset uses anonymized Customer IDs rather than real personal information.
- No authentication-gated data or secrets are required to run the public demo.
- No API keys or credentials are committed to the repository.

11 Personal Reflection

Key Learnings

- Translating raw clustering output into business-friendly labels (Bronze/Silver/Gold) is just as important as the modeling itself — stakeholders act on labels, not cluster numbers.
- Simple, well-chosen visualizations (a heatmap, a trend line) often communicate more than a complex model.
- Getting a Streamlit app onto a public URL end-to-end taught me the full path from notebook to deployed product, not just model building in isolation.

Industry Best Practices Applied

- Clear separation between data exploration (notebook) and the presentation layer (dashboard).
- A README that documents the dataset, methodology, and results for anyone reviewing the repo.
- Reproducible setup via a documented list of dependencies.

Future Roadmap

- Replace the static CSV with a connection to a live database or API for real-time updates.
- Add cross-validation and a richer test set to validate the churn model beyond a single train/test split.
- Introduce demand forecasting (e.g., Prophet) to complement the current descriptive analysis with predictive lead time for inventory decisions.
- Add basic monitoring (e.g., data drift checks) if this were to move toward production use.
- Re-validate all models on a real-world transactional dataset (e.g., UCI Online Retail) with genuine noise and repeat-purchase behavior, to obtain production-representative accuracy figures.

“This project reflects a complete journey from raw retail data to a deployed, decision-ready analytics product — and a foundation I plan to keep building on.”

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